

Aparna Balaji

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EDUCATION

University of Texas at Dallas

Bachelor of Science in Computer Science

Dallas, TX

Expected May 2029

University of Ottawa

Honors Bachelor of Science in Computer Science

Ottawa, ON

September 2025 – May 2026

- GPA: 3.9/4.0
- Gooi Seong Chneh and Gooi Seong Gum Scholarship
- Relevant Coursework: Linear Algebra I, Intro to Computing I + II, Digital Systems, Discrete Math for Computing

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, TypeScript, JavaScript

Frameworks and Tools: React, Git, Vercel, Docker, Arduino IDE, Microsoft Office Suite, Canva

Spoken Languages: Native Fluency in Tamil; Elementary Fluency in Spanish

WORK EXPERIENCE

iCode

McKinney, TX

Instructor

February 2024 – October 2024

- Instructed coding fundamentals to approximately 30 students per week, ages 5 to 12, in Scratch, Python, Roblox Studio, and animation software, through both one-on-one and group sessions. Created over 5 reusable lesson plans that other instructors adopted.
- Directed students to complete original projects, including animated videos, Roblox games, and simple digital instruments built with conductive materials.

ACADEMIC & PERSONAL PROJECTS

Market Trading Simulator | *Python*

September 2026 – Present

- Built a systematic trading and backtesting engine from scratch in Python, including buy and sell logic, portfolio tracking, and realized versus unrealized profit and loss calculations.
- Implemented a walk-forward testing loop with a look-ahead guard to prevent data leakage, and coded momentum and mean-reversion trading strategies to test against historical data.

Pet Adoption Bio Generator | *React, Claude API, Vercel*

July 2026

- Built a web application that generates written pet adoption bios by integrating the Claude API, designing and implementing both the React frontend and the backend independently.
- Deployed the project on Vercel as a self-directed portfolio piece demonstrating full API integration.

ESP32 Digital Camera and Embedded System | *ESP32, C++, Arduino, OV2640, MicroSD*

February 2026 – Present

- Built a custom digital camera using an ESP32 microcontroller, capable of capturing and saving multiple photos successfully.
- Developed the embedded firmware controlling image capture and storage on the device.

IEEE uOttawa Student Branch Website | *React, Payload CMS, Kanban, Git, Vercel*

November 2025 – May 2026

- Collaborated with the IEEE uOttawa software team to build and deploy the branch website using Next.js, Payload CMS, and Tailwind CSS.
- Built reusable React components and integrated Payload CMS so non-technical officers can update site content without editing code, and the site is currently being used by IEEE uOttawa.

LEADERSHIP

IEEE uOttawa Student Branch

Ottawa, ON

First Year Coordinator

September 2025 – May 2026

- Co-organized *Bits, Bytes, and Breadboards*, a hands-on Arduino and circuitry workshop for around 45 first-year students, serving as lead organizer over two months of planning. The event won the IEEE uOttawa Award for Premier Technical Event of the Year.
- Elected VP Academic of the branch in April 2026 prior to transferring.

Computer Science Student Association

Ottawa, ON

Junior Vice President Academic

September 2025 – May 2026

PERSONAL ACCOMPLISHMENTS

IEEE uOttawa Award for Premier Technical Event of the Year, IEEE uOttawa Student Branch

March 2026

Co-Author, “Determining the Separation and Position Angle of WDS 04078-6903,” JDSO 17(2), 145–149

2021

Co-Author, “Observations of Small-Separation Red Dwarf Binary Pairs Using Bispectrum Phase Reconstruction,” JDSO 18(3), 285–295

2022